

Ian Kolaja

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Summary

PhD in ML systems and large-scale scientific computing with hands-on experience building production-grade data pipelines and distributed ML infrastructure. Proven ability to work across the full ML lifecycle from data representation and feature engineering through model training, evaluation, and deployment in safety-critical, research-driven environments. Hands-on experience with document processing, retrieval-augmented generation, language model integration, and AI analysis workflows.

Education

University of California, Berkeley

August 2021 to August 2025

PhD, Nuclear Engineering

3.63

Designated Emphasis in Computational and Data Science and Engineering

Highlighted Courses: Applications of Parallel Computers, Graduate Nuclear Reactions and Interactions of Radiation with Matter, Nuclear Security: The Nexus Between Policy and Technology, Advanced Nuclear Reactors, Special Topics in Environmental Aspects of Nuclear Energy: Thermal Hydraulics, Applied Machine Learning, Advanced Concepts in Radiation Detection and Measurements, Physical Principles of CT, PET, and SPECT Imaging, Introduction to Statistical Computing, Concepts of Probability

University of California, Berkeley

August 2017 to December 2019

B.S., Nuclear Engineering

3.79

Highlighted Courses: Nuclear Criticality Safety, STEM Communication, Radiation Biophysics and Dosimetry, Design for Advanced Nuclear Systems

Fullerton College

August 2015 to June 2017

A.S., Mathematics and Chemistry

3.76

Work Experience

University of California, Berkeley

August 2021 to August 2025

Graduate Student Researcher

- Designed and implemented an end-to-end ML pipeline for forecasting high-dimensional time-series data, including data preprocessing, feature engineering, encoding, model training, evaluation, and integration into automated control workflows. Deployed TensorFlow/Keras models on distributed HPC infrastructure and achieved $R^2 = 0.991$ on held-out data.
- Designed and implemented a nuclear data analysis system featuring data processing pipelines, statistical evaluation frameworks, and an interactive Flask application. Integrated LLM-based agents for automated analysis and tool orchestration, and built a RAG pipeline for semantic retrieval of relevant experimental reports.
- Imputed missing experimental uncertainties in nuclear data records using NLP: extracted sentence-level embeddings from experimental report PDFs using SciBERT, computed cosine similarity against prototype phrases to generate data quality feature vectors, and applied KNN to identify and transfer uncertainty values from similar but complete reports.
- Built scalable Python data pipelines for large-scale simulation and ML workflows on distributed HPC clusters, enabling efficient processing of high-dimensional datasets and supporting rapid iteration on ML models.
- Inventor on provisional patent for a novel optics-driven technique to rapidly assess nuclear fuel.

University of California, Berkeley

January 2022 to May 2022

Graduate Student Instructor (GSI)

- Taught Introduction to Nuclear Reactor Theory. Received a 4.94 out of 5.0 on various student-graded metrics including teaching effectiveness, helpfulness in understanding course material, and preparedness.
- Held hybrid discussion sections going over course material in greater detail with an emphasis of helping students understand how to utilize content covered in lectures. Organized review sessions and held office hours to ensure students were prepared.

Kairos Power

June 2019 to September 2021

Core Design Engineer

- Led development of a production-grade Python simulation platform integrating high-fidelity Monte Carlo models, enabling large-scale parameter studies and providing a reusable framework for downstream data processing and analysis workflows.
- Created a Python framework for generating detailed reactor models from simple object definitions. Automated the generation of LaTeX technical reports to ensure consistent documentation of model details.

Lawrence Berkeley National Lab

January 2018 to December 2019

Undergraduate Researcher

- Determined a physically justifiable neutron flux spectrum for a fast neutron inelastic scattering experiment conducted at the Baghdad Nuclear Research Institution in 1978. Performed sensitivity analysis on the fitted flux function parameter in order to determine its uncertainty. Ongoing work will allow for the data coupled with the flux spectrum to serve as a nuclear model performance benchmark.
- Assessed the performance of the neutron flux spectrum by using it in a comparison between the experimental data and cross section calculations performed by TALYS, a nuclear reaction model.
- Developed code for the automatic analysis of gamma spectra that can identify and fit peaks using radioactive decay data.

Utilized code to rule out the presence of fission products in the analysis of actinium-225, a coveted medical isotope.

- Acquired data for detection of super heavy element moscovium-288 using the Berkeley Gas-filled Separator (BGS) coupled with the 88-inch cyclotron. Utilized LabVIEW software to monitor detection events and adjust BGS parameters over time.

UC Berkeley Thermal Hydraulics Lab

November 2018 to December 2019

Undergraduate Researcher

- Developed software in MATLAB and LabVIEW allowing a MIDI controller to be used as an analog controller for the Compact Integral Effects Test (CIET) facility.
- Developed an electronic procedure module in LabVIEW that guides reactor operators through experiments in CIET. Designed a data structure that can be easily created by other researchers to dynamically populate the electronic procedure with human readable instructions. The module is responsive to CIET's state, automatically advancing through the procedure as certain physical conditions are met.
- Designed procedures with graduate student for human factors experiment intended to test the effectiveness of different operator support tools. Conducted survey to supplement experimenter observations with participant's user feedback.

UC Berkeley

August 2018 to December 2018

Course Reader

- Graded homework assignments and examinations for Nuclear Reactions and Radiation course at UC Berkeley. Contributed to the development of assignment and exam content using insight as a student. Hosted office hours to answer student questions about course material and grading policy.

Community Service

JOYA Scholars

September 2015 to May 2017

Tutor and Mentor

- Tutored ten high school students from families of low-income communities to help a population that historically has had little access to higher education excel academically. Helped students with math, computer science, and biology.
- Mentored five middle school students in career selection, college admission, and overcoming academic challenges. Inspired one student to pursue aerospace engineering as a major and apply to Stanford University.

Fullerton College Associated Students

September 2015 to February 2016

Student Senator

- Served as vice chair of Education and Curriculum Committee. Planned events augmenting student success. Among most volunteer hours served in Senate.
- Served on the district's Council on Budget and Facilities and Men and Women of Distinction Committee. Assisted in drafting of resolutions advocating for students who become homeless during the semester and increasing health fees to improve mental health resources on campus.

Skills

- **ML & Statistics:** TensorFlow, Keras, scikit-learn, neural networks (CNNs, RNNs/LSTMs), deep learning, time-series modeling, data encoding, pretrained language models, transformer embeddings, cosine similarity, KNN, PCA, optimization, uncertainty quantification, regression analysis, hypothesis testing
- **Programming Languages:** Python, MATLAB, C++, R, Lua
- **Software Engineering & Programming Practices:** algorithm design, computational optimization, complexity analysis, dynamic programming, unit testing, CI/CD, Git
- **Systems & Infrastructure:** HPC clusters, distributed computing, SLURM, Python multiprocessing, CUDA
- **LLM & AI Systems:** HuggingFace, Retrieval-augmented generation, LLM tool orchestration, function calling, structured message schemas, conversation state management (Claude / Anthropic API)
- **Web & Visualization:** HTML, CSS, JavaScript, Flask, Apache2, Matplotlib, Bokeh, Mayavi
- **Scientific Modeling & Simulation:** Serpent, MCNP, TALYS, SHADOW3, GADRAS, RELAP5, ORIGEN
- **Instrumentation:** Gamma Spectroscopy, LabVIEW
- **Documentation & Communications:** Microsoft Office Suite, LaTeX, Overleaf, Speech Writing, Technical Writing, Social Media Outreach
- **Operating Systems:** Linux, Windows, MacOS
- **Creative & Media Tools:** Adobe Premiere Pro, Adobe Photoshop, Music Production, Ableton, Music Theory

Publications & Preprints

- I. Kolaja et al., "Predicting and forecasting reactivity and flux using long short-term memory models in pebble bed reactors during run-in," *arXiv*, November 2025, doi: <https://arxiv.org/abs/2511.05118>
- I. Kolaja et al., "Burnup measurement using bent crystal diffraction spectrometers for pebble bed reactors," *Annals of Nuclear Energy*, vol. 233, August 2026
- I. Kolaja et al., "Machine Learning Prediction of Pebble History in Pebble Bed Reactors," *Proceedings of the International Conference on Physics of Reactors (PHYSOR 2024)*, pp. 1437-1446, April 2024
- N. Satval et al., "Neutronics, thermal-hydraulics, and multi-physics benchmark models for a generic pebble-bed fluoride-salt-cooled high temperature reactor (FHR)," *Nuclear Engineering and Design*, vol. 384, no. C, November 2021

Awards

Best Paper (Reactor Physics)

December 2025

Berkeley Nuclear Engineering

Awarded for submitted journal article "Burnup measurement using bent crystal diffraction spectrometers for pebble bed

reactors." Selected from reactor physics papers submitted by UC Berkeley NE grad students in 2025.

Outstanding GSI

March 2023

UC Berkeley Graduate Division Teaching & Resource Center

Awarded in recognition for exceptional achievements as a teacher. About ten percent of GSIs teaching on UC Berkeley's campus receive this award.

Wootton Award for Superior Service and Leadership

June 2020

Berkeley Nuclear Engineering

Awarded in recognition of superior service and leadership to the Department of Nuclear Engineering

Chancellor's Fellowship for Graduate Study

March 2020

Berkeley Nuclear Engineering

Highly competitive campuswide fellowship nominated by department faculty. Funded two years of PhD study.

U.S. Nuclear Regulatory Commission Scholarship

May 2018

NRC

Awarded by NRC for commitment to the field and remaining above a 3.0 GPA.

Student of Distinction

May 2017

Fullerton College

Selected among twenty students recognized for academic achievement in a school of twenty-five thousand students by committee of faculty and students. Invited to audition to speak at commencement, became one of four finalists.

Carbon Contributions Challenge Finalist

March 2017

MIT Solve

Proposed a solution with team for Massachusetts Institute of Technology (MIT) Solve's Carbon Contributions Challenge and was selected as one of eleven finalists. Solutions were accepted from around the world with many finalists being graduate students from top tier schools. Designed a process called "Direct Methane Conversion" which allows for the collection, filtration, and conversion of methane into graphene and hydrogen. Research was presented at the United Nations headquarters to global thought leaders, entrepreneurs, and innovators. Answered questions from a panel of judges selected from industry and academia.

Artist of the Year Finalist in Film

April 2015

Orange County Register

Nominated by high school video teacher. Submitted "Holding Back the Tide" and became one of ten finalists. Interviewed by a panel of professional filmmakers and instructors from colleges such as University of California and Chapman.